

FORMATION OF HIGHLY PERMEABLE SOIL ON A HEAVY SOIL BASE UNDER TIDAL IRRIGATION AND DRAINAGE CONDITIONS[†]

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ABSTRACT

Abadan date palm plantations have long been irrigated by rivers under the tidal effects of the Persian Gulf. During high tides, river water makes subirrigation possible while during low tides the same ditches drain out the plant root zone. Recently parts of the plantation have been under traditional drainage and surprisingly failed to work efficiently. In order to find the cause, the hydraulic conductivity of the soil layers was measured using parallel drains which show that the topsoil up to 70 cm is very permeable, but then up to 95 cm it is impermeable. Piezometric measurements also show that the soil between 70 and 130 cm is impermeable. The ratio of contribution (RC) of the soil layers to the total flow in the soil profile decreases with depth. Values of the RC for the first 60 and 70 cm soil depths are about 91 and 98% respectively. Although soil texture in the first layer has been classified as clay, biological activities, mostly live roots, have changed both the magnitude and size of the pores. It was concluded that in tidal subirrigation, only the upper part of the soil which has been under the root spread can be assumed to be permeable. Copyright © 2018 John Wiley & Sons, Ltd.

KEY WORDS: date plantation; drain; hydraulic conductivity; piezometer; recharge ditch

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RÉSUMÉ

Les plantations de palmiers dattiers d'Abadan ont longtemps été irriguées par des rivières sous contrôle aval des marées du golfe Persique. Pendant les marées hautes, l'eau de la rivière rend la sub-irrigation possible via les fossés tandis qu'en marée basse, les mêmes fossés drainent la zone racinaire. Récemment, des parties de la plantation ont été réaménagées avec du drainage par tuyaux enterrés et, étonnamment, ces zones n'ont pas réussi à fonctionner efficacement. Afin de trouver la cause, les conductivités hydrauliques des couches de sol ont été mesurées en utilisant des drains parallèles, qui montrent que le sol superficiel jusqu'à 70 cm est très perméable, puis devient progressivement imperméable à partir de 95 cm. Les mesures piézométriques montrent également que le sol entre 70 et 130 cm est imperméable. Le rapport de la contribution (RC) des couches de sol au débit total dans le profil de sol diminue avec la profondeur avec des valeurs variant de 91.8 et 98.5% pour les premiers 60 et 70 cm de profondeur, respectivement. Bien que la texture du sol de la première couche ait été classée en argile, les activités biologiques, principalement les racines vivantes, ont modifié à la fois l'ampleur et la taille des pores. Il a été conclu que dans la sub-irrigation des marées, seule la partie supérieure du sol où les racines se sont propagées peut être supposée perméable et que les drains profonds posés en dessous, dans le niveau imperméable, ne peuvent pas fonctionner correctement. Copyright © 2018 John Wiley & Sons, Ltd.

MOTS CLÉS: plantation de dattes; drainer; conductivité hydraulique; piézomètre; fossé de recharge

INTRODUCTION

Abadan Island developed as a tidal mudflat formed by sedimentary deposition in the delta of the Karun and Arvand

river systems. In the long past, continuous and gradual sedimentation during regularly occurring flooding, along with the tidal effects of the sea, eventually raised the mudflat to

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[†]Formation of a highly permeable soil on a heavy soil base under the effect of tidal irrigation and drainage conditions.

a level where vegetation grew. The total date palm plantation area on Abadan Island is around 28 000 ha. The width of the plantation strips is between 1 and 3 km along the Arvand and Bahmanshir rivers. Tidal irrigation is the natural subirrigation of levee soils in coastal plains with river water under tidal influence. It applies in (semi) arid zones at the mouth of a large river estuary or delta where a considerable tidal range is present. The subirrigation is effected by digging tidal canals from the river shore into the mainland that will guide the river water inland at high tides. At low tide, the canals and the soil drain out again, which promotes the aeration of the soil. The field ditches with a depth of 70–80 cm and spacing of 10–12 m, depending on the permeability of the topsoil, receive their water from lateral ditches and tidal canals. Permeability of the topsoil should be high enough to allow the soil between parallel field ditches to become almost saturated during around 6 h between high tide and normal water level, while it could drain out once in nearly 6 h between normal water level and low tide. Proper and efficient working of the system in recent decades indicates the sustainability of the system.

Due to rising river water salinity caused by reduction of river flows, parts of the plantations were equipped with a subsurface corrugated pipe drainage system in the last decade. But surprisingly, the subsurface drainage system failed to work efficiently in some parts, for reasons unknown. In this area, the subsurface drainage system consists of a perforated plastic pipe (diam. 10 cm) wrapped in synthetic material installed at a depth of 90–130 cm and with lateral spacing of 30 m. In the bare soils of Abadan Island, where there is no evidence of vegetation, hydraulic properties of the soils are very weak. Regarding soil permeability, under tidal conditions, fluctuation of water in the canals and ditches has created suitable conditions for vegetation in the neighbouring lands between the tidal canals and ditches. Because of spatial variations in soil structure due to uneven distribution of biopores in the soil profile as well as its diversity along the field, point measurements of soil permeability have given different results. Furthermore, macroporosity (such as interpedal voids, root and worm channels, or cracks) and its spatial distribution has a significant effect on saturated water flow and solute transport (Bouma, 1991). This feature is generally not easily quantified through analyses of point measurements representing a small soil volume. So, for better understanding of biopore development in the soil profile and consequently permeability of the soil at each depth increment, it was required to consider a large volume of soil in order to integrate the uneven distribution of biopores in the whole body of the soil profile. The purpose of this research was to make the new drainage developments in tidal areas more effective, if possible. The hypothesis of this research is that drainage of the soils in this area is constrained by the impermeability of the subsoil.

Saturated hydraulic conductivity (K_s value) in the soil profile can be highly variable from place to place and can also be variable at different depths, which means spatial variability. Because of many factors such as uneven distribution of plant roots and agricultural activities, hydraulic conductivity values can be variable not only in connection with different soil layers, but also within one soil layer. Since roots cannot stay alive in a saturated zone, their spread as well as their activities are limited to the topsoil, hence there is a significant difference between the hydraulic conductivity of the topsoil and other layers. The saturated hydraulic conductivity of Abadan Island soil has been investigated by many researchers but there is still a doubt about the location of impermeable layer in the soil profile (Ashrafi *et al.*, 2015).

There are many methods for measuring hydraulic conductivity of soils. So far most of the measurements applied here have been based on small-scale field methods which may not be suitable for strongly layered soils or for soils with irregular pore space distribution. The auger hole method was the sole method used in previous studies. So, investigators have reached an average hydraulic conductivity which could be very different from the hydraulic conductivity of each layer.

Large-scale field methods guarantee representative K values, where the problem of variation is eliminated as much as possible. Large-scale field methods are rather expensive and time-consuming compared to small-scale methods. The advantage of large-scale determinations is that the flow paths of the groundwater and the natural irregularities of the K values along these paths are automatically taken into account in the overall K value found with the method. It is then not necessary to determine the variations in the K values from place to place, in the horizontal and vertical directions, and the overall K value found can be used directly as input into drainage formulae. A second advantage is that the variation in K values found is considerably less than that found with small-scale methods (Oosterbaan and Nijland, 1994). Of the different large-scale field methods of hydraulic conductivity measurement, parallel drains are more accurate as well as being representative of the whole area of groundwater flow to the drains (Ritzema *et al.*, 2006).

Preferential flow may have a drastic effect on the hydraulic conductivity of the soil and severely limits the applicability of standard flow models. Models are mostly based on Richards' equation (Gerke, 2006). The pore network, especially pore size distribution and connectivity, is believed to control soil hydraulic properties. Preferential flow pathways in this intact structured soil consist of a complex network of earthworm burrows, root channels, interaggregate macropores, and mesopores or even micropores in the soil matrix (Luo *et al.*, 2008). Permeability problems in irrigated soils may be alleviated by root systems that increase water flow by creating macropores. However, decaying

roots of alfalfa produced stable macropores, while wheat produced none such (Mitchell *et al.*, 1995). Therefore, it depends on the shape and pattern of the roots. Root systems create networks of preferential flow and thus influence water pressures in soils which may alter subsurface flow. Conceptual examples are presented to illustrate how root architecture and properties (e.g. diameter, length, orientation, topology, sinuosity and decay rate) affect the creation of root channels and thus affect preferential flow (Ghosem *et al.*, 2011). Furthermore, Barges Tobella *et al.* (2014) demonstrated that trees have a positive impact on soil hydraulic properties, influencing groundwater recharge.

MATERIALS AND METHODS

In this study the saturated hydraulic conductivity of the different soil layers was measured using auger hole and inverse auger hole methods at 16 stations. At each site, a trench was dug to approximately 2 m depth to study the soil profile. Profile no. 2 located in the experimental field was the representative profile of the region. Soil colour and layering were examined in the field and soil properties determined in the laboratory. Soil sampling was done with an auger and soil properties determined at the soil laboratory of the Agricultural Engineering Research Institute (AERI) in Karaj (Tajik *et al.*, 2015).

Three points were chosen next to each profile in order to measure the saturated hydraulic conductivity of the soil layers, each with three replications at depths of 50, 100, 150 and 200 cm approximately. Results obtained from the study have shown that even soil layers at a specific depth

at each station had different values of saturated hydraulic conductivity. Comparison of the value of hydraulic conductivity at different sites in the same soil layer showed significant differences as well (Ashrafi *et al.*, 2015). The existence of earthworm burrows and root channels creates preferential flow pathways in the soil profile, and the auger hole crossing these flow pathways could be the reason for altering the real value of saturated hydraulic conductivity of the soil layers. Thus, the experiments continued in the research date palm field of Shalheh Haj Hossein village near the Arvand River (approximately 500 m far from the river), with a large-scale field method called the parallel drains method in order to obtain more accurate and reliable values of saturated hydraulic conductivity of the soil layers.

The experimental field was isolated on two sides by two open deep drains and on the other two sides by using polyethylene plastic rolls to a depth of around 1.5 m. The irrigation water was recharged by a ditch laid in the middle of the two drains. Inflow and outflow of the middle ditch were measured by WSC (Washington State College) type II flumes. In the method used in this research, a system of open drains was placed on relatively horizontal very low permeable layers. The available drain in the experimental field was excavated to a depth of 100 cm. The width and depth of the recharge ditch were 40 and 100 cm, respectively, with a length of 16 m which is equal to the length of the drains. Drain spacing (L) was 12 m, free water level in the drains was h_2 and maximum groundwater table at the recharge ditch was h_1 , as shown in Figure 1. The origin of the x - y axis was selected to be in the middle of the bed of the recharge ditch. The horizontal axis is positive to the right direction which extends on the very low permeable,

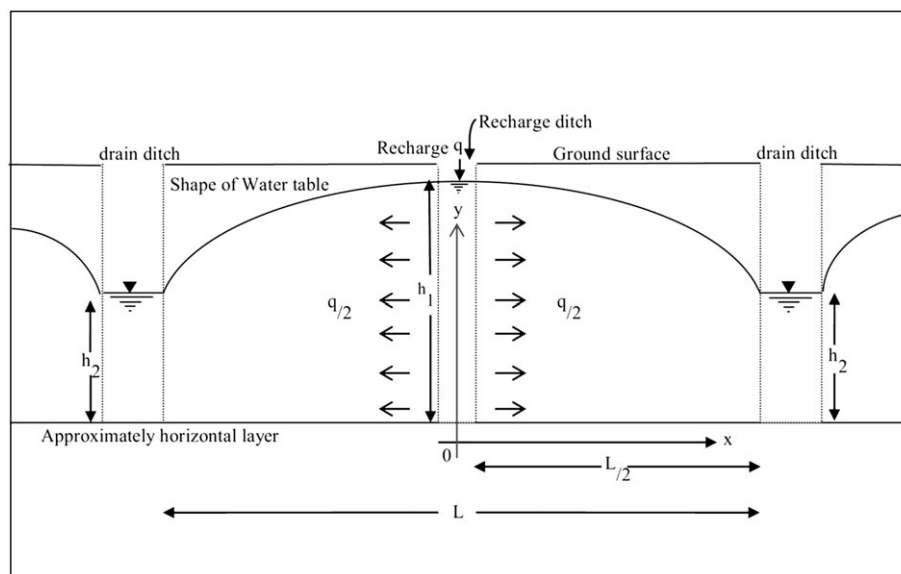


Figure 1. Schematic view of the steady-state flow to the open ditches, placed on a relatively horizontal layer.

approximately horizontal layer. The vertical axis is positive to the upward direction.

Since the process of steady-state drainage flow is symmetrical around y-axis, it is placed midway between the drains. Based on Darcy's law (1856) and Dupuit–Forcheheimer's assumptions, Equation 1 was used for the calculation of saturated hydraulic conductivity of the soil at different water levels of the drains:

$$q = \frac{(h_1^2 - h_2^2)}{L} * K \quad (1)$$

A small watertight dam was constructed at the lower end of each drain as shown in Figure 2. In order to maintain the required water level at each drain, 10 PVC pipes (approximately 0.7 m long and 10 cm in diameter) were installed horizontally in the body of the dams with a 10 cm incremental step from the bottom to the top of the dams. To measure the groundwater table in the soil profile, six observation



Figure 2. Small watertight dams at the lower ends of the drains. [Colour figure can be viewed at wileyonlinelibrary.com]

wells on the right and six on the left side of the recharge ditch (with 1 m horizontal distance between each other) were placed at a depth of 130 cm at the midpoint of the field. The experimental field also was equipped with four piezometer batteries on the right and four on the left side of the recharge ditch at the midpoint of the field with a distance of 1.2 m between each other. The depth of each piezometer battery in the soil was 40, 50, 60, 70, 90 and 130 cm-surface. The horizontal distance of each piezometer in the piezometer batteries was 25 cm. The piezometer batteries were installed parallel to the recharge ditch and drains.

The initial groundwater table in the observation wells and piezometers was measured before starting the experiment. The required level of the water table for measuring the saturated hydraulic conductivity of each soil layer was maintained by plugging the lower PVC pipes at the body of the dams. The experiment for each level of the water table (at the drains) was started by recharging the middle ditch. Inflow and outflow from the recharge ditch were measured by WSC flumes. Each experiment continued until it reached an equilibrium condition. This condition was obtained when the water level or outflow from the lower-end flume of the recharge ditch became constant. The water table in the recharge ditch was kept at a constant level in all experiments and different levels of water table at the drains were examined. A steady-state flow condition between recharge ditch and drains in each experiment for a required water table at the drains lasted almost 5 h.

RESULTS AND DISCUSSION

Soil description

The description of the profile excavated adjacent to the experimental field is given in Table I.

The results showed that the soil had three distinct layers. The first brown layer with less than 80 cm thickness is distinguished by plant root and biological activity. The second 100 cm layer has redoximorphic properties with visible mottling signs and slaking behaviour. The third layer is permanently saturated and has a green to grey colour, and behaves as sticky gley. The soils of Abadan are mostly similar and are classified as Typic Aquicambids (United States Department of Agriculture–Natural Resources Conservation Service (USDA–NRCS), 2014) but a better method of classification is the World Reference Base for Soil Resource (Food and Agriculture Organization of the United Nations (FAO), 2014), which provides the best definition of the characteristics and behaviour of Abadan soils. Considering this classification, the soils could be classified as Tidalic Gleysols (Tajik *et al.*, 2015).

Table I. Soil description of profile 2, located adjacent to the experimental field.

Depth (cm)	Soil description
0–20	Clayey texture with live roots, brown colour
20–75	Porous clayey texture with many live and thick roots and macroorganisms, brown colour
75–100	Clayey texture with a few gley signs, few brown mottles, few live roots, hard digging
100–140	Clayey texture with a few gley signs, few brown and black mottles, thin live roots, and fine sand lode
140–160	Clayey texture with a few gley signs, many brown and black mottles, few live roots
160–180	Clayey texture with many gley signs and many brown mottles, few roots
180–200	Silty clay texture with many gley signs, brown mottles and few roots, green to greyish brown colour

Determination of the location of impermeable layer in the soil profile

In the first step of the experiment, it was required to determine the location of impermeable layer in the soil profile. For this purpose, the water table in the drains was kept at the lowest elevation until subsurface drain flow reached the steady-state condition. Results of this experiment showed that the maximum rate of recharge happens when the water table in the drains is at the lowest level (96.5 cm-surface). This rate of recharge remained constant for other water table levels of 81 and 68 cm-surface. From a depth of 68 cm, as the elevation of the water table in the drains increased the value of recharge decreased, as shown in Figure 3.

Piezometric study

Piezometers were used for measuring the potential of water in different soil layers. In order to have a completely saturated condition of the soil profile, from top to deep soil in the experimental field, the highest water table in the drains

was used for piezometric study. As experiments were conducted continuously every day from early morning to evening, the soil layers had approximately 12 h to discharge water which had remained from the previous experiment. Before starting the experiment, the initial condition of the groundwater table, which was measured in the 12 observation wells, was more or less at the same elevation of 145 cm, as shown in Figures 4(a)–(d). At the end of the experiment, the final condition of the groundwater table in the soil profile (between drains) varied between depths of 21 and 6 cm-surface in the vicinity of the drain and recharge ditch, respectively, as shown in Figures 4(a)–(d). Results obtained for different depths of piezometers, 40, 50, 60 and 70 cm, are shown in Figures 4(a), (b), (c) and (d) respectively. Almost all the initial and final conditions of the groundwater table in the piezometers installed at depths of 90 and 130 cm did not change during the experimental time.

The initial condition of the groundwater table in the piezometers installed at depths of 40 and 50 cm showed that there is no saturated condition at those depths. At the end of the experiments, the groundwater table in these piezometers was approximately identical to levels of the groundwater table in the soil profile at each point, as shown in Figures 4(a) and (b). The initial condition of groundwater tables in the 60 cm piezometers showed that, out of 8 piezometers, 3 still had water from the last experiment (12 h before), but the other 5 were empty. Elevation of the groundwater table in these three piezometers had not significantly changed at the end of the experiment, as shown in Figure 4(c). The other five piezometers at this depth were receiving water from the soil during the experiment. The water level in these piezometers were almost lower than the groundwater table in the soil profile as shown in Figure 4(c).

The initial condition of the groundwater table in the 70 cm piezometers showed that all eight piezometers still had water at the average depth of 65 cm which had remained from the last experiment (12 h before), but the final condition of the

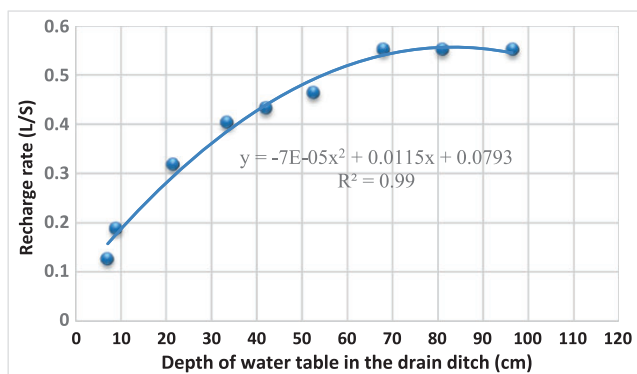


Figure 3. Relationship between the recharge and depth of the water table in the drains. [Colour figure can be viewed at wileyonlinelibrary.com]

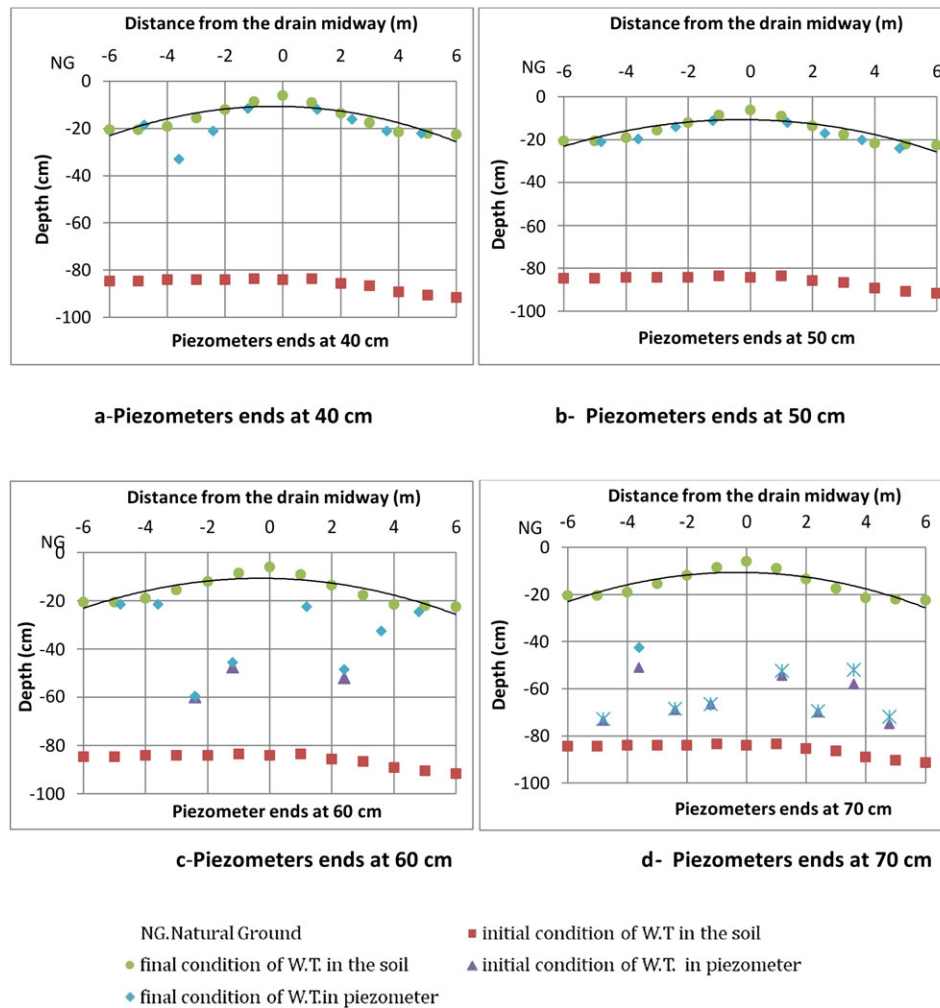


Figure 4. Groundwater table in the soil profile and piezometers installed at depths of 40, 50, 60 and 70 cm. [Colour figure can be viewed at wileyonlinelibrary.com]

groundwater table in these piezometers had not significantly changed at the end of the experiment. Although the average groundwater table in the soil profile at the end of the experiment was 14 cm-surface, the average depth of water table in these piezometers had remained at 62 cm-surface at the same time, as shown in Figure 4(d). Comparing the average elevation of the groundwater table in the soil profile with the final elevation of the water table in the piezometers installed at a depth of 70 cm indicates that movement of water from the soil to these piezometers is very slow. The average change of water table in these piezometers was only 3 cm during the experimental time. This indicates that compared to upper soil layers, the permeability of the soil at a depth of 70 cm is very low. Almost all the initial and final conditions of the water table in the piezometers installed at depths of 90 and 130 cm did not change during the experimental time. This means that the soil layer located between 90 and 130 cm depths is not permeable at all.

Hydraulic conductivity calculation

Results obtained from measurements of recharge flow in this study have shown that the soil layer which is located between depths of 68 and 96.5 cm is not permeable. Also, results which were taken from piezometric study have shown that impermeability of the soil in the experimental field starts from 70 cm-surface and extends to 130 cm-surface in the soil profile. Equation (1) was used for hydraulic conductivity calculation based on the boundary condition of the flow at the recharge ditch and drains. Then, the value of the hydraulic conductivity of the soil at each depth increment in the soil profile was calculated using primary K -values and concerning flow thicknesses. The saturated hydraulic conductivity of the soil at the midpoint of each depth increment is shown in Figure 5. As shown in this figure, saturated hydraulic conductivity of the soil at depths of 15, 27, 37, 46 and 55 cm are 35,

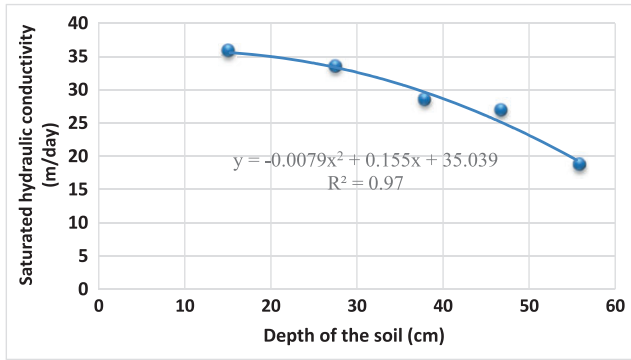


Figure 5. Relationship between depth of the soil and its hydraulic conductivity (based on field measurements). [Colour figure can be viewed at wileyonlinelibrary.com]

33, 28, 26 and 18 m day⁻¹, respectively. The equation obtained from the fitted line to the measured data (Equation 2) was used and saturated hydraulic conductivity of the soil at different depths was calculated. Based on Equation 2, saturated hydraulic conductivity of the soil at depth 77 cm becomes very low and negligible:

$$K = 0.0079 * Z^2 + 0.155 * Z + 35.0 \quad (2)$$

Ratio of contribution of each layer to total flow in the soil profile

Based on the results obtained from piezometric study and hydraulic conductivity measurements, there are only 70–75 cm of permeable soil in the experimental field and beyond 75 cm depth, soil should be considered impermeable. Moreover, results obtained from the soil profile description in this study have shown most of the plant roots and bioactivity in the soil profile are located in the first 75 cm of soil depth and also, there is an inverse relation between soil depth and bioactivity in the soil profile. As depth increases, bioactivity decreases in the soil profile (Table I). As porosity in the soil has a direct effect on the movement of water in the soil profile and the value of saturated hydraulic conductivity is the consequence of the contribution of the pores to convey a volume of water in the soil, so Equation 2 can be used for determination of the percentage contribution of the biopore at each soil layer to the total flow in the soil profile. According to Equation 2, we have saturated hydraulic conductivity of the soil for any depth in the soil profile. So, for a given soil layer, located between depths of z_1 and z_2 , the average saturated hydraulic conductivity K_{avr} in this soil layer can be calculated as follows:

$$K_{avr} = \frac{1}{(z_2 - z_1)} \int_{z_1}^{z_2} k dz \quad (3)$$

According to Darcy's law, for a given hydraulic gradient, i , the amount of water that moves within the z_1 and z_2 layer in

the unit length of the field is defined as follows:

$$Q = K_{avr} * (z_2 - z_1) * i \quad (4)$$

By substituting k_{avr} from Equation 3 to Equation 4 we have.

$$Q = \left[\frac{1}{(z_2 - z_1)} \int_{z_1}^{z_2} K dz \right] * (z_2 - z_1) * i \quad (5)$$

By substituting k from Equation 2 to Equation 5 and taking integrals, we have.

$$Q = \left[\int_{z_1}^{z_2} (-0.0079 * Z^2 + 0.155 * Z + 35.039) dZ \right] * i \quad (6)$$

$$Q = [-0.00263 * Z^3 + 0.0775 * Z^2 + 35.039 * Z]_{z_1}^{z_2} * i \quad (7)$$

In real conditions, the water table at the recharge ditch for the all field experiments was at 6 cm-surface, and maximum discharge (recharge) from the soil profile happened when the water table in the drains was at 68 cm-surface. According to the soil description in this study, most of the live roots and bioactivity are located in the first 75 cm of soil depth, and also based on Equation 2, the saturated hydraulic conductivity of the soil becomes negligible at a depth of 77 cm, so for the calculation of maximum flow in the soil profile, we assume an upper limit of the permeable layer in the field experiment to be at a depth of 77 cm. So for a given hydraulic gradient, i , the ratio of the flow between the depths of z_1 and z_2 to the maximum flow in the soil profile is defined as follows:

$$Rc_{z_1, z_2} = \frac{q_{z_1, z_2}}{Q_{6, 77}} = \frac{\left[\int_{z_1}^{z_2} K_{ave} * dZ \right] * i}{\left[\int_6^{77} K_{ave} * dZ \right] * i} \quad (8)$$

When the elevation of the water table at the drains is equal to or less than the elevation of the impermeable layer, the hydraulic gradient, i , in Equation 8, becomes equal to 1. So, we can use the following equation in order to calculate the ratio of the contribution (RC) of each depth increment to the total flow in the soil profile. The results obtained from the RC calculation for each depth increment are shown in Table II.

$$RC_{z_1, z_2} = \frac{[-0.00263 * Z^3 + 0.0775 * Z^2 + 35.1 * Z]_{z_1}^{z_2}}{[-0.00263 * Z^3 + 0.0775 * Z^2 + 35.1 * Z]_6^{77}} \quad (9)$$

As shown in Table II, the ratio of contribution of each 5 cm depth increment decreases with depth. As mentioned above, the lower and upper limits (z_1 and z_2) of the RC

Table II. The ratio of the contribution (RC) of each depth increment to total flow in the soil profile.

z_1 (cm)	z_2 (cm)	RC (%)	Cumulative RC (%)	Cumulative of each quarter RC (%)
6	10	8.2	8.2	38.5
10	15	10.2	18.4	
15	20	10.1	28.6	
20	25	9.9	38.5	
25	30	9.6	48.0	27.2
30	35	9.1	57.1	
35	40	8.5	65.6	
40	45	7.8	73.5	
45	50	7.1	80.5	26.1
50	55	6.1	86.7	
55	60	5.1	91.8	
60	65	4.0	95.8	
65	70	2.7	98.5	8.2
70	75	1.4	99.9	
75	77	0.1	100	

calculation were considered to be at 6 and 77 cm-surface. So, if the thickness of the soil between the upper and lower limits is divided into four equal zones, the thickness of each zone becomes 17.8 cm. As shown in Table II, the RC of each quarter to the total flow in the soil profile from the top to the bottom is 38.5, 27.2, 26.1 and 8.2% respectively. The first quarter has the highest contribution and the last quarter the lowest contribution to the total flow in the soil profile. As mentioned in the piezometric study of this research, based on our measurements, the problem of impermeability in the soil profile starts at a depth of 70 cm. As shown in Table II, soil at a depth of 70 cm and more only makes a 1.51% contribution to the total flow in the profile. Also, as shown in Figure 4(c), out of 8 piezometers at the depth of 60 cm, 3 did not receive water and the water table in the other 5 piezometers was less than the groundwater table at the soil profile (Figure 4c). As shown in Table II, the contribution of the soil depths of 60 cm and more to total flow is only 8.2%. The functions of piezometers at the depth of 60 cm and the value of RC for depths of 60 cm and more indicate that when the remaining cumulative RC in the depth increment is less than 10%, the problem of impermeability starts in the soil profile. These comparisons between the piezometric study and RC calculation indicate a good correlation between the piezometric study and RC calculation in this study.

As mentioned in the flow measurement section of this study, the contribution of soil located at depths of 68 cm and more to the drain discharge is negligible and the soil was specified as impermeable. The cumulative RC calculated for this depth shows a value of 97.5%. This means that there is only less than 2.5% flow at depths of 68 cm and more in the soil profile. In this case also, the results of RC

calculation and flow measurement in the field completely match each other.

In the case of soil profile description, as presented in Table I, most of the roots and bioactivity are located in the first 75 cm soil depth. The effect of roots on permeability of the soil has been studied by many researchers. In this regard, according to the study conducted by Logsdon and Allmaras (1991), roots are able to form well-connected macropores or channels and also normally grow into rigid pores broader than their own diameters. Other researchers (Aber *et al.*, 1985; Li and Ghodrati, 1994; Jørgensen *et al.*, 2002; Bogner *et al.*, 2010; Germann *et al.*, 2012) believe that channels created by plant roots may contribute to water and solute transport, especially macropore flow or preferential flow. In this regard, Jørgensen *et al.* (2002) found that 94% of flow in a clay-rich till was conducted along root channels, while only 6% flowed along fractures without root channels. The RC calculation in the Abadan soil also indicated that the RC of each quarter of the plant root zone (6–77 cm) to total flow in the soil profile is 38.5, 27.2, 26.1 and 8.2% respectively. It is well known that the abundance and distribution of plant roots in the soil profile decrease with depth. The RC calculated for each 5 cm depth increment showed that the RC of the soil increment to total flow also decreases with depth, as presented in Table II.

CONCLUSION

The soil profile description in this study has shown that, of the three layers distinguished, most of the biological activities were concentrated in the first layer the thickness of which is less than 80 cm. Although the soil texture in the first layer has been classified as a clayey texture in the first 75 cm layer, biological activities, mostly live and thick roots, have changed the magnitude of the pores. Based on the soil texture classification, the saturated hydraulic conductivity of a soil with clayey texture should be less than 0.2 m day^{-1} (Ritzema *et al.*, 2006), but values obtained from the line fitted to the field data show that the saturated hydraulic conductivity changes from 35 to 0.135 m day^{-1} at depths of 5–77 cm respectively. The value of the saturated hydraulic conductivity of the experimental field shows that, in the first 60 cm layer, the soil is very permeable. This diversity of saturated hydraulic conductivity can be interpreted only by addressing the description of the soil profile in Table I. As presented there, live and thick roots and macroorganisms (all biological activities) are concentrated in the first 75 cm depth of the experimental soil. The ratio of the contribution (RC) of the first 60 and 75 cm soil depths to total flow in the soil profile is about 91.8 and 99.9% respectively. This means that the first 75 cm soil depth, where all bioactivities are concentrated, is

responsible for transmitting almost all water in the soil profile. As mentioned by different researchers such as Mitchell *et al.* (1995) and Luo *et al.* (2008), preferential flow pathways consist of a complex network of earthworm burrows, root channels, interaggregate macropores, and mesopores or even micropores in the soil matrix that increase the water flow of the soil. Also decaying roots of different plants form preferential pathways and consequently cause high hydraulic conductivity of such a soil.

From the viewpoint of soil permeability, under tidal conditions, the fluctuation of water in canals and ditches has created suitable conditions for vegetation on adjacent land located between the tidal canals and ditches. The penetration of plant roots and other bioactivity are the main reason behind the formation of a highly permeable layer in the top layer of the soil profile, which had not been considered by researchers in this area till now.

Finally, from the viewpoint of implementation of a subsurface pipe drainage system, special care should be taken in a fragile soil environment, because soil permeability in this area is the consequence of bioactivity in the soil profile. The soil permeability also was controlled by the stability of the groundwater table in the soil profile. As the problem of impermeability starts from a depth of 70–75 cm, implementation of pipe drains beyond this depth is not suggested; also, installing pipe drains at 75 cm depth runs a high risk of roots entering the drainpipes.

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